सी एस आई आर – राष्ट्रीय भौतिक प्रयोगशाला
CSIR-NATIONAL PHYSICAL LABORATORY
(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद्)
(Council of Scientific and Industrial Research)
(राष्ट्रीय मापिकी संस्थान (एनएमआई), सदस्य बीआईपीएस एवं हस्ताक्षरकर्ता सीआईपीएम - एमआरए)
(National Metrology Institute (NMI), Member BIPM and Signatory CIPM - MRA)
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अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE

A Density Hydrometer

प्रमाण पत्र संख्या / Certificate No.

25080664/D1.01/C-425

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pages
04.11.2025	04.11.2028	1	1

1. Calibrated for : Beatus Social Compliance Services Ltd.
Kamardanga Road, Ichapur,
Howrah - 711 104 (West Bengal), India

Customer's Ref. No. : BSCSL/NPL/25-26/0130, Dated : 19.08.2025

2. Description and identification of instrument/item : A Density Hydrometer
Range : (0.700 to 0.750) g/cm³at 15°C
Resolution : 0.001g/cm³
Make : Leimco
Type : M-50, LST (Pet.)
Serial No. : 427712, BSCSL-PT-020-2026 (Loop-A)

3. Environmental conditions : Temperature : (25 ± 3) °C
Relative humidity : (50 ± 10) %

4. Standard(s) used with associated measurement uncertainty : Standard hydrometers Sub-Div. # 1.01/D/8784, 8890 and 8785 and liquid of appropriate density and associated uncertainty ± 0.0004 g/cm³

5. Metrological traceability of standard(s) used : The standard(s) used for measurement (is) are traceable to the National Standard which realize the units of mass according to the International System of Units (SI).

6. Principle/Methodology of calibration And calibration procedure number : CSIR-NPL Calibration Procedure No.Sub-Div.#1.01/Doc.3/CP#DH/D-02 is used in which the hydrometer under calibration is compared in a liquid of appropriate density

7. Results :

Scale reading at 15°C (g/cm³)	Correction to be applied at 15°C (g/cm³)	Uncertainty of measurement (g/cm³)
0.700 0.720 0.740 0.750	+ 0.000 3 + 0.000 4 + 0.000 5 + 0.000 3	± 0.0009

The reported uncertainty is at coverage factor $k = 2$ which corresponds to a coverage probability of approximately 95 % for a normal distribution.

8. Date(s) of calibration : 31.10.2025

9. Remarks :
1. Calibrated only the scale of this hydrometer.
2. The correction has to be algebraically added to the density hydrometer reading in order to get actual observed value.

अंशांकनकर्ता: Calibrated by: (D C SHARMA)

जाँचकर्ता: Checked by: (MANOJ KUMAR)

प्रभारी वैज्ञानिक: Scientist-in-charge: (NIDHI SINGH)

जारीकर्ता: Issued by: डॉ. पी. नीवास राव रागम
Director's Nominee

06/11/2025